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Inventor(s)

KOMAZAWA; Hisao

INFORMATION PROCESSING SYSTEM, NON-TRANSITORY COMPUTER READABLE MEDIUM STORING INFORMATION PROCESSING PROGRAM, AND INFORMATION PROCESSING METHOD

Abstract

An information processing system includes a processor configured to perform continuous printing of plural printed materials while inspecting printing quality, in a case where a printed material having an inspection result of fail is generated, stop the continuous printing of the printed materials and reprint the failed printed material, and in a case where an inspection result of the reprinted printed material is pass, print a subsequent printed material following the passed reprinted material.

Inventors: KOMAZAWA; Hisao (Kanagawa, JP)

Applicant: FUJIFILM Business Innovation Corp. (Tokyo, JP)

Family ID: 1000008143396

Assignee: FUJIFILM Business Innovation Corp. (Tokyo, JP)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is based on and claims priority under 35 USC 119 from Japanese Patent Application No. 2024-035288 filed Mar. 7, 2024.

BACKGROUND

(i) Technical Field

[0002] The present invention relates to an information processing system, a non-transitory computer readable medium storing an information processing program, and an information processing method.

(ii) Related Art

[0003] JP2021-165020A discloses an inspection system that inspects the quality of a printed material, the inspection system including an inspection unit that inspects the quality of the printed material based on a comparison between a scan image obtained by scanning the printed material and a standard image, a display control unit that displays a screen for receiving a first mode in which a discharge destination of the printed material is not switched regardless of an inspection result of the printed material, or a second mode in which the discharge destination of the printed material is switched according to the inspection result of the printed material, a reception unit that receives the first mode in which the discharge destination of the printed material is not switched regardless of the inspection result of the printed material, or the second mode in which the discharge destination of the printed material is switched according to the inspection result of the printed material, via the screen, and a control unit that controls the discharge destination of the printed material based on the inspection result obtained by the inspection performed by the inspection unit and the mode received by the reception unit.

[0004] JP2022-146528A discloses an inspection device that inspects a printed material by comparing a standard image with a printed material that has been printed, the inspection device including a control unit that exclusively controls, as the inspection mode of the inspection, a mode for switching the paper discharge destination of a printed material in which a defect is found, and an inspection job for inspecting a printed material, with a plurality of printed materials as one set, or an inspection type for inspecting printed materials having an order.

[0005] JP2022-148533A discloses a printing system including a printing unit that performs a printing process based on a print job, on a page basis, and an inspection unit that inspects the presence or absence of a print defect on a printed sheet by comparing a read image obtained by reading an image on the printed sheet of a page printed based on the print job with a normal image registered in association with the print job, in which the print job includes setting information for an operation mode that specifies the printing operation in a case where a print defect is detected in the inspection, and the printing unit determines which page of the pages specified in the print job is to be printed next, in a case where the print defect is detected in the inspection, according to the operation mode indicated by the setting information.

SUMMARY

[0006] Aspects of non-limiting embodiments of the present disclosure relate to an information processing system, a non-transitory computer readable medium storing an information processing

program, and an information processing method that reduce the amount of the printed material to be discarded, in a case of performing continuous printing of a plurality of printed materials while the printing quality is inspected, and a printed material having an inspection result of fail is generated, as compared with a case where the continuous printing is resumed from the failed printed material.

[0007] Aspects of certain non-limiting embodiments of the present disclosure address the above advantages and/or other advantages not described above. However, aspects of the non-limiting embodiments are not required to address the advantages described above, and aspects of the non-limiting embodiments of the present disclosure may not address advantages described above.

[0008] According to an aspect of the present invention, there is provided an information processing system including: a processor configured to: perform continuous printing of a plurality of printed materials while inspecting printing quality; in a case where a printed material having an inspection result of fail is generated, stop the continuous printing of the printed materials and reprint the failed printed material; and in a case where an inspection result of the reprinted printed material is pass, print a subsequent printed material following the passed reprinted material.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] Exemplary embodiment(s) of the present invention will be described in detail based on the following figures, wherein:

[0010] FIG. 1 is a diagram showing a schematic configuration of an image forming system according to an exemplary embodiment;

[0011] FIG. 2 is a diagram illustrating an operation of the image forming system shown in FIG. 1;

[0012] FIG. 3 is a diagram illustrating an operation of the image forming system shown in FIG. 1;

[0013] FIG. 4 is a block diagram showing a hardware configuration of the image forming system shown in FIG. 1;

[0014] FIG. 5 is a functional block diagram of the image forming system shown in FIG. 1;

[0015] FIG. 6 is a diagram illustrating an operation of the image forming system shown in FIG. 1;

[0016] FIG. 7 is a diagram illustrating an operation of the image forming system shown in FIG. 1;

[0017] FIG. 8 is a diagram illustrating an operation of the image forming system shown in FIG. 1; and

[0018] FIG. 9 is a flowchart showing a flow of a print inspection process of the image forming system shown in FIG. 1.

DETAILED DESCRIPTION

[0019] Hereinafter, an exemplary embodiment will be described with reference to the drawings.

Overview of Image Forming System

[0020] FIG. 1 shows an image forming system **10** according to an exemplary embodiment. The image forming system **10** includes a printing device **20**, an inspection device **30**, and a post-processing device **40**. The image forming system **10** is an example of an information processing system and an information processing apparatus.

Printing Device

[0021] The printing device **20** includes a plurality of accommodation trays **20T** for accommodating paper. The printing device **20** has a printing function of printing an image on paper supplied from any accommodation tray **20T**, among the plurality of accommodation trays **20T**. The paper is an example of a recording medium, and the accommodation tray **20T** is an example of an accommodation unit.

Inspection Device

[0022] The inspection device **30** has an inspection function of receiving the paper on which the

image is printed from the printing device **20** and inspecting the printing quality of the received printed paper (printed material). The inspection device **30** determines whether or not there is an abnormality such as corner folding, contamination of printing, or misalignment in the printed material received from the printing device **20**.

[0023] Specifically, the inspection device **30** includes a purge tray **30T** that discharges the printed material. The inspection device **30** receives a standard image (normal image), such as a rasterized image, generated on the basis of image data to be printed on the paper, from the printing device **20**. In addition, the inspection device **30** scans the image of the printed material received from the printing device **20**. Then, the inspection device **30** collates the standard image received from the printing device **20** with an inspection image such as a rasterized image based on the scanned image and determines whether or not there is an abnormality in the printed material.

[0024] As a result, in a case where it is determined that there is an abnormality in the printed material, the inspection device **30** sets the inspection result to fail and discharges the printed material to the purge tray **30T**. On the other hand, in a case where it is determined that there is no abnormality in the printed material, the inspection device **30** supplies the printed material into the post-processing device **40** by setting the inspection result as pass. The purge tray **30T** is an example of a fail discharge unit.

Post-Processing Device

[0025] The post-processing device **40** executes various types of post-processes, such as a folding process, a stapling process, a punch hole process, and a booklet making process, on the printed material inspected by the inspection device **30**, and discharges the post-processed printed material into the predetermined discharge tray **40T**. The processing contents of the post-processing device **40** can be changed as appropriate. In addition, the discharge tray **40T** is an example of a pass discharge unit.

Continuous Printing

[0026] The image forming system **10** has a continuous printing function. In this continuous printing, for example, a plurality of continuous printed materials are continuously printed while the printing quality is inspected. Specifically, in the continuous printing, the printing device **20** prints the image on each of the plurality of sheets of paper supplied in sequence from the accommodation tray **20T**, and the inspection device **30** sequentially inspects the printed material on which the image is printed. In the inspection device **30**, the printed material having the inspection result of pass is subjected to post-processing in the post-processing device **40** and is discharged into the predetermined discharge tray **40T**.

[0027] Here, as shown in FIG. 2, during continuous printing, in a case where a printed material having an inspection result by the inspection device **30** of fail is generated (printed material of **6p**), not only the failed printed material but also the printed materials (printed materials of **7p** and **8p**) being printed following the failed printed material, that is, all the sheets of paper on the paper transport path from the accommodation tray **20T** to the inspection by the inspection device **30** are discharged to the purge tray **30T**.

[0028] Thereafter, as shown in FIG. 3, the continuous printing (reprinting) is resumed from the printed material (printed material of **6p**) having the inspection result by the inspection device **30** of fail. Then, as shown by the two-dot chain line in FIG. 3, the reprinted material (printed material of **6p**) that has passed the quality inspection by the inspection device **30** and the subsequent printed materials (printed materials of **7p** to **11p**) following the reprinted material are discharged to the discharge tray **40T** of the post-processing device **40**. In a case where a printed material having an inspection result by the inspection device **30** of fail is generated, only the failed printed material (printed material of **6p**) is discharged to the purge tray **30T**, and in a case where the printed materials (printed materials of **7p** and **8p**) being printed following the failed printed material are discharged to the discharge tray **40T** of the post-processing device **40**, the page order of the printed materials is shifted in the discharge tray **40T**.

[0029] However, as described above, in a case where a printed material having an inspection result of fail is generated during continuous printing, and not only the failed printed material but also all the printed materials being printed following the failed printed material are discharged to the purge tray **30T**, the amount of paper to be discarded (amount of waste paper) is increased as compared with a case where only the failed printed material is discharged to the purge tray **30T**.

[0030] In particular, since the printed material that is determined to fail by the inspection device **30** may be determined to fail again by the inspection device **30** at the time of reprinting, the amount of paper to be discarded may increase.

[0031] In addition, in a case where a printed material having an inspection result of fail is generated in the inspection device **30** as the paper transport path from the accommodation tray **20T** to the inspection device **30** is longer, the number of sheets of paper discharged to the purge tray **30T** is increased.

[0032] Further, in a case where a printed material having an inspection result of fail is generated in the inspection device **30** as the transport speed of the paper increases, the number of sheets of paper remaining in the paper transport path from the accommodation tray **20T** to the inspection device **30** increases, so that the number of sheets of paper discharged to the purge tray **30T** increases. In addition, in a case where a printed material having an inspection result of fail is generated in the inspection device **30**, not only the amount of paper to be discarded but also the amount of toner used for printing and the printing time are increased.

[0033] Therefore, in the image forming system **10** according to the present exemplary embodiment, in the inspection device **30**, in a case where a printed material having an inspection result of fail is generated, the continuous printing is stopped, and only the failed printed material is reprinted. In the inspection device **30**, in a case where the inspection result of the reprinted printed material is pass, a subsequent printed material following the passed reprinted material is printed. Hereinafter, the image forming system **10** according to the present exemplary embodiment will be described in detail.

Hardware Configuration of Image Forming System

[0034] First, a hardware configuration of the image forming system **10** will be described. It should be noted that, in the following, for convenience of description, a hardware configuration of the image forming system **10** will be described, but the printing device **20**, the inspection device **30**, and the post-processing device **40** may separately include the hardware configuration as shown in FIG. 4.

[0035] As shown in FIG. 4, the image forming system **10** has components including a central processing unit (CPU) **50**, a read only memory (ROM) **52**, a random access memory (RAM) **54**, a storage **56**, an input unit **58**, a display unit **60**, a communication interface (I/F) **62**, an image reading unit **64**, an image forming unit **66**, an inspection unit **68**, and a post-processing unit **70**. The configurations are connected to each other via a bus **72** in a communicable manner.

[0036] The CPU **50** is a central processing unit and executes various programs or controls each unit. That is, the CPU **50** reads out a program from the ROM **52** or the storage **56** and executes the program using the RAM **54** as a work area. The CPU **50** controls each of the above-described configurations and performs various types of arithmetic processing according to the program recorded on the ROM **52** or the storage **56**. In addition, the CPU **50** is an example of a processor.

[0037] The ROM **52** stores various information processing programs and various types of data. The RAM **54** temporarily stores the program or the data as the work area of the CPU **50**.

[0038] The storage **56** is configured with a hard disk drive (HDD) or a solid state drive (SSD), and stores various programs including an operating system and various types of data. In addition, the storage **56** stores a response processing program.

[0039] The input unit **58** includes an operation button or the like and is used to perform various inputs. The display unit **60** is, for example, a liquid crystal display and displays various types of information. The display unit **60** may be a touch panel type and may function as the input unit.

[0040] The communication I/F **62** is an interface for communication with other apparatuses. For example, a standard, such as Ethernet (Registered Trademark), FDDI, or Wi-Fi (Registered Trademark), is used.

[0041] The image reading unit **64** includes a scanner or the like and optically reads image data from a document or the like. The image forming unit **66** prints the image data read by the image reading unit **64** or the image data received from an operation terminal (not shown) on a recording medium such as paper.

[0042] The inspection unit **68** includes a scanner or the like, and optically reads image data from the paper (printed material) on which the image is printed by the image forming unit **66**.

[0043] The post-processing unit **70** includes various post-processing mechanisms such as a folding processing mechanism of a recording medium such as paper, a stapling processing mechanism, a punch hole processing mechanism, and a booklet manufacturing processing mechanism, and performs a post-process on the printed material inspected by the inspection device **30**.

Function of Image Forming System

[0044] Next, functions of the image forming system **10** will be described.

[0045] In a case where the information processing program described above is executed, the image forming system **10** implements various functions of the printing device **20**, the inspection device **30**, and the post-processing device **40** by using the above hardware resources.

Printing Device

[0046] Specifically, as shown in FIG. 5, the printing device **20** functionally includes a print request acquisition unit **22**, a supply unit **24**, a printing unit **26**, and a printing control unit **28**.

[0047] The print request acquisition unit **22** acquires, for example, information (print data) of the print image input to the image forming system **10** or a print request (print job) such as a paper size and the number of printed sheets of paper.

[0048] The supply unit **24** supplies, to the printing unit **26**, the paper accommodated in the predetermined accommodation tray **20T** among the plurality of accommodation trays **20T**. The printing unit **26** prints an image on the paper supplied from the supply unit **24**.

Continuous Printing

[0049] The printing control unit **28** controls the overall operation of the printing device **20**, and executes continuous printing in which an image is printed on each of a plurality of sheets of paper accommodated in the accommodation tray **20T** and the plurality of printed materials are sequentially supplied to the inspection device **30**.

[0050] Specifically, the printing control unit **28** controls the supply unit **24** based on the print request acquired by the print request acquisition unit **22** to sequentially supply the plurality of sheets of paper from the predetermined accommodation tray **20T** to the printing unit **26** among the plurality of accommodation trays **20T**. In addition, the printing control unit **28** transmits the print data to be printed on the plurality of sheets of paper to the printing unit **26**. Then, the printing control unit **28** causes the printing unit **26** to print the image based on the print data on each of the plurality of sheets of paper supplied from the accommodation tray **20T**, and sequentially supplies the plurality of printed materials from the printing unit **26** to the inspection device **30**.

[0051] In addition, in the continuous printing, the printing control unit **28** transmits the print data for performing printing on each of the plurality of sheets of paper to an inspection control unit **39** in the inspection device **30** described below.

Reprinting

[0052] For example, as shown in FIG. 2, in a case where in the inspection device **30**, a printed material (printed material of **6p**) having an inspection result of fail is generated, the printing control unit **28** stops the continuous printing of the printed material and executes reprinting of only the printed material (printed material of **6p**) having an inspection result of fail.

[0053] Specifically, as shown in FIG. 6, the printing control unit **28** controls the supply unit **24** based on the inspection result received from the inspection control unit **39** described below, and

supplies one sheet of paper from a predetermined accommodation tray **20T** among the plurality of accommodation trays **20T** to the printing unit **26**. Then, the printing control unit **28** causes the printing unit **26** to reprint the image of which the inspection result is fail on one sheet of paper supplied from the accommodation tray **20T**, and supplies the reprinted material (reprinted material of **6p**) that is reprinted, from the printing unit **26** to the inspection device **30**.

[0054] In a case where the inspection result of the reprinted material by the inspection device **30** is passed, the printing control unit **28** resumes the continuous printing of the subsequent printed materials following the reprinted material.

Test Printing

[0055] In a case where the inspection result of the reprinted material (reprinted material of **6p**) is fail in the inspection device **30** described below, the printing control unit **28** executes test printing (blank paper printing) in which no image is printed on the paper accommodated in the accommodation tray **20T** and the paper is supplied to the inspection device **30**. The test printing is defined as pseudo-printing in which an image is not printed on paper, and is used to determine whether the cause of the fail is the paper itself or the printing device **20** in a case where the inspection result of the inspection device **30** is fail.

[0056] Specifically, as shown in FIG. 7, in the test printing, the printing control unit **28** controls the supply unit **24** to supply the paper from the predetermined accommodation tray **20T**, among the plurality of accommodation trays **20T** to the printing unit **26**. In addition, the printing control unit **28** transmits test data (blank paper data) having no print data to the printing unit **26**, and causes the printing unit **26** to not print an image on the paper supplied from the accommodation tray **20T** but to supply the paper as a test printed material to the inspection device **30**. The test printed material is inspected by the inspection device **30** and then discharged into the purge tray **30T**.

[0057] In addition, in the test printing, the printing control unit **28** transmits information on the test image (blank paper image) having no print data to the inspection control unit **39** in the inspection device **30** described later.

Retest Printing

[0058] In the test printing of the test printed material, in a case where the inspection result of the test printed material is fail, the printing control unit **28** executes retest printing of switching the accommodation tray **20T** and performing the test printing again in the inspection device **30** described below.

[0059] Specifically, as shown in FIG. 8, in the retest printing, in a case where there is another accommodation tray **20T** which is different from the one accommodation tray **20T** used for the test printed material having the inspection result of fail and in which paper of the same size and the same type as the paper accommodated in the one accommodation tray **20T** is accommodated, the printing control unit **28** controls the supply unit **24** to supply the paper from the other accommodation tray **20T** to the printing unit **26**.

[0060] In addition, in the same manner as in the test printing, the printing control unit **28** causes the printing unit **26** to not print an image on the paper supplied from the accommodation tray **20T** but to supply the paper as a retest printed material to the inspection device **30**. The retest printed material is inspected by the inspection device **30** and then discharged into the purge tray **30T**.

[0061] For convenience of description, the second and subsequent test printings are referred to as retest printings, and the test printed materials printed by the second and subsequent test printings are all referred to as retest printed materials.

Inspection Device

[0062] The inspection device **30** functionally includes an inspection image acquisition unit **32**, a standard image acquisition unit **34**, a quality inspection unit **36**, a discharge destination switching unit **38**, and an inspection control unit **39**.

[0063] The inspection image acquisition unit **32** scans an image from a printed material printed by the printing unit **26**, and generates an inspection image such as a rasterized image based on the

scanned image.

[0064] The standard image acquisition unit **34** receives the print data or the test data transmitted from the printing control unit **28** as the standard image.

[0065] The quality inspection unit **36** compares the inspection image read by the inspection image acquisition unit **32** with the standard image received by the standard image acquisition unit **34** to determine whether or not there is an abnormality in the printed material, that is, whether the printing quality of the printed material is pass or fail.

[0066] Specifically, the quality inspection unit **36** compares the inspection image with a standard image to determine whether or not there is an abnormality such as a corner folding, printing contamination, or misalignment. In a case where it is determined that there is no abnormality in the printed material, the quality inspection unit **36** determines the printed material as pass. In a case where it is determined that there is an abnormality in the printed material, the quality inspection unit **36** determines the printed material as fail.

[0067] The discharge destination switching unit **38** switches the discharge destination of the printed material inspected by the quality inspection unit **36** between the post-processing device **40** and the purge tray **30T**.

[0068] The inspection control unit **39** controls the operation of the entire inspection device **30**. Specifically, the inspection control unit **39** causes the quality inspection unit **36** to inspect the printing quality of the printed material or the reprinted material. In a case where the printing quality of the printed material or the reprinted material is passed by the quality inspection unit **36**, the inspection control unit **39** causes the discharge destination switching unit **38** to switch the discharge destination of the printed material or the reprinted material to the post-processing device **40**.

[0069] On the other hand, in a case where the inspection result of the printed material or the reprinted material by the quality inspection unit **36** is fail, the inspection control unit **39** switches the discharge destination of the printed material or the reprinted material to the purge tray **30T** by the discharge destination switching unit **38**.

[0070] Further, the inspection control unit **39** causes the quality inspection unit **36** to inspect the printing quality of the test printed material or the retest printed material. The inspection control unit **39** switches the test printed material or the retest printed material is switched to the purge tray **30T** by the discharge destination switching unit **38**.

[0071] In addition, the inspection control unit **39** transmits information such as the inspection result by the quality inspection unit **36** and the number of pages (page numbers) of the printed materials determined to be failed to the printing control unit **28**.

Post-Processing Device

[0072] The post-processing device **40** functionally includes a post-processing control unit **42**. The post-processing control unit **42** controls the operation of the entire post-processing device **40**. Specifically, the post-processing control unit **42** controls various post-processing mechanisms, such as a folding processing mechanism, a stapling processing mechanism, a punch hole processing mechanism, and a booklet making processing mechanism, and performs a post-process on the printed material supplied from the inspection device **30**.

[0073] The printing control unit **28**, the inspection control unit **39**, and the post-processing control unit **42** control the overall operation of the image forming system **10** in cooperation with each other by transmitting and receiving information to and from each other.

Printing Inspection Process of Image Forming System

[0074] Next, a printing inspection process by the image forming system **10** will be described.

[0075] In the image forming system **10**, in a case where continuous printing of a plurality of printed materials is executed and the printed material printed by the printing device **20** is supplied to the inspection device **30**, the print inspection process shown in FIG. **9** is executed in the image forming system **10**. The printing inspection process is an example of a printing inspection processing method.

[0076] First, in step **S10**, the CPU **50** inspects the printing quality of the printed material, the reprinted material, the test printed material, or the retest printed material.

[0077] Next, in step **S12**, the CPU **50** determines whether the inspection result of the printing quality is pass or fail. Then, in a case where the inspection result is fail, the CPU **50** transitions to step **S14**.

[0078] Next, in step **S14**, the CPU **50** determines whether or not the inspection target is the printed material. Then, in a case where the CPU **50** determines that the inspection target is the printed material, the CPU **50** transitions to step **S16**.

[0079] Next, in step **S16**, the CPU **50** stops the continuous printing.

[0080] Next, in step **S18**, the CPU **50** discharges the printed material having the inspection result of fail and all the printed materials being printed following the printed material to the purge tray **30T**.

[0081] Next, in step **S20**, the CPU **50** executes the reprinting of printing only the printed material having the inspection result of fail and ends the process.

[0082] On the other hand, in step **S14**, in a case where the CPU **50** determines that the inspection target is not the printed material, the CPU **50** transitions to step **S22**.

[0083] Next, in step **S22**, the CPU **50** determines whether or not the inspection target is the reprinted material. Then, in a case where the CPU **50** determines that the inspection target is the reprinted material, the CPU **50** transitions to step **S24**.

[0084] Next, in step **S24**, the CPU **50** executes test printing (blank paper printing) of not printing an image on the paper and ends the process.

[0085] In the first test printing, the inspection device **30** is supplied with paper from the accommodation tray **20T** used for the printed material and the reprinted material having the inspection result of fail via the printing device **20**. That is, the CPU **50** executes test printing on the paper accommodated in the same accommodation tray **20T** as the accommodation tray **20T** of the paper used for the printed material and the reprinted material having the inspection result of fail. In this case, in the printing device **20**, the image is not printed on the paper.

[0086] On the other hand, in step **S22**, in a case where the CPU **50** determines that the inspection target is not the reprinted material, that is, the inspection target is the test printed material or the retest printed material, the CPU **50** transitions to step **S26**.

[0087] Next, in step **S26**, the CPU **50** determines whether or not there is another switchable accommodation tray **20T** among the plurality of accommodation trays **20T**.

[0088] Specifically, the CPU **50** determines whether or not there is another accommodation tray **20T** which is different from the one accommodation tray **20T** used for the test printed material having the inspection result of fail and in which paper having the same size and of the same type as the paper accommodated in the one accommodation tray **20T** is accommodated. Then, in a case where the CPU **50** determines that there is another switchable accommodation tray **20T**, the CPU **50** transitions to step **S28**.

[0089] Next, in step **S28**, the CPU **50** switches the accommodation tray used for the retest printing from one accommodation tray **20T** to another accommodation tray **20T**, and transitions to step **S24**. In step **S24**, the CPU **50** executes the retest printing.

[0090] On the other hand, in step **S26**, in a case where the CPU **50** determines that there is no other switchable accommodation tray **20T**, the CPU **50** transitions to step **S34** which will be described later. In this step **S26**, in a case where the inspection result of the test printed material is fail, the retest printing is performed until there are no more switchable accommodation trays **20T**.

[0091] In addition, in step **S12** described above, in a case where the CPU **50** determines that the inspection result of the printing quality is pass, the CPU **50** transitions to step **S30**.

[0092] Next, in step **S30**, the CPU **50** determines whether or not the inspection target is the printed material. Then, in a case where the CPU **50** determines that the inspection target is the printed material, the process ends.

[0093] On the other hand, in step **S30**, in a case where the CPU **50** determines that the inspection

target is not the printed material, the CPU 50 transitions to step S32.

[0094] Next, in step S32, the CPU 50 determines whether or not the inspection target is the test printed material. Then, in a case where the CPU 50 determines that the inspection target is the test printed material, the CPU 50 transitions to step S34.

[0095] Next, in step S34, the CPU 50 cancels the printing, and ends the process. This is because in a case where the inspection result of the test printed material is pass, the paper is not abnormal, and there is a high possibility that the cause of the fail of the inspection result of the printed material and the reprinted material is in the printing device 20.

[0096] On the other hand, in step S32, in a case where the CPU 50 determines that the inspection target is not the test printed material, that is, the inspection target is the reprinted material or the retest printed material, the CPU 50 transitions to step S36.

[0097] Next, in step S36, the CPU 50 resumes continuous printing from the printed material having the inspection result of fail and ends the process. That is, the CPU 50 performs continuous printing of the printed material having an inspection result of fail and subsequent printed materials following the printed material, and ends the process.

[0098] In a case where the inspection target is the retest printed material in step S32, the CPU 50 resumes continuous printing on the paper accommodated in another accommodation tray 20T switched from the one accommodation tray 20T in step S36.

Modification Example

[0099] Next, modification examples of the above-described exemplary embodiment will be described.

[0100] In the above-described exemplary embodiment, in the retest printing, the one accommodation tray 20T used for the test printed material having the inspection result by the inspection device 30 of fail is switched to another accommodation tray 20T in which paper having the same size and the same type as the paper accommodated in the one accommodation tray 20T is accommodated. However, in the retest printing, the one accommodation tray 20T used for the test printed material having the inspection result of fail may be switched to another accommodation tray 20T in which paper having a different size or a different type from the paper accommodated in the one accommodation tray 20T is accommodated. In addition, the test printing may be omitted.

[0101] In addition, in the above-described exemplary embodiment, in the test printing, the accommodation tray 20T is not switched to another accommodation tray 20T, and the same accommodation tray 20T as the accommodation tray 20T of the paper used for the printed material having the inspection result of fail is used. That is, in the above-described exemplary embodiment, test printing is performed on the paper accommodated in the same accommodation tray 20T as the accommodation tray 20T of the paper used for the printed material having the inspection result of fail. However, in the test printing, the accommodation tray 20T may be switched to another accommodation tray 20T. In addition, the retest printing may be omitted.

[0102] In addition, in the above-described exemplary embodiment, in a case where a printed material having the inspection result by the inspection device 30 is fail is generated, test printing may be performed after repeating the reprinting a predetermined number of times.

[0103] In addition, in the above-described exemplary embodiment, in a case where the inspection result of the reprinted material by the inspection device 30 is pass, the reprinting of the subsequent printed materials following the reprinted material is resumed. However, in a case where the inspection result of the reprinted material by the inspection device 30 is pass, for example, the inspection result by the inspection device 30 may be determined each time one subsequent printed material is printed, and in a case where a predetermined number of the inspection results are continuous, the subsequent printed materials may be switched to continuous printing.

[0104] In the embodiments above, the term “processor” refers to hardware in a broad sense. Examples of the processor include general processors (e.g., CPU: Central Processing Unit) and dedicated processors (e.g., GPU: Graphics Processing Unit, ASIC: Application Specific Integrated

Circuit, FPGA: Field Programmable Gate Array, and programmable logic device).

[0105] In the embodiments above, the term “processor” is broad enough to encompass one processor or plural processors in collaboration which are located physically apart from each other but may work cooperatively. The order of operations of the processor is not limited to one described in the embodiments above, and may be changed.

[0106] In the above exemplary embodiment, the mode in which each program is installed in the ROM or the storage has been described, but the present invention is not limited to this. Each program according to the above exemplary embodiment may be provided in the form of a recording on a computer readable storage medium. For example, each program according to the above exemplary embodiment may be provided in the form of a recording on an optical disc such as a compact disc (CD)-ROM and a digital versatile disc (DVD)-ROM or in the form of a recording on a semiconductor memory such as a universal serial bus (USB) memory and a memory card. Further, each program according to the above-described exemplary embodiment may be acquired from an external apparatus via a communication I/F.

[0107] In addition, in the exemplary embodiment described above, a case where the processing in the image forming system **10** is implemented by a software configuration by using a computer by executing a program is described, but the present disclosure is not limited thereto. For example, the processing in the image forming system **10** may be implemented by a hardware configuration or a combination of the hardware configuration and the software configuration.

[0108] Further, needless to say, the configuration of the image forming system **10** described in the exemplary embodiment described above is an example, and an unnecessary part may be deleted or a new part may be added without departing from the gist of the present disclosure.

[0109] In addition, needless to say, the flow of the processes in the image forming system **10** described in the above exemplary embodiment is also an example, and an unnecessary step may be deleted, a new step may be added, or processing order may be changed without departing from the gist of the present disclosure.

[0110] The following supplementary notes are further disclosed regarding the above-described exemplary embodiment. [0111] (((1)))

[0112] An information processing system comprising:

[0113] a processor configured to: [0114] perform continuous printing of a plurality of printed materials while inspecting printing quality; [0115] in a case where a printed material having an inspection result of fail is generated, stop the continuous printing of the printed materials and reprint the failed printed material; and [0116] in a case where an inspection result of the reprinted printed material is pass, print a subsequent printed material following the passed reprinted material.

[0117] (((2)))

[0118] The information processing system according to (((1))), wherein the processor is configured to: [0119] perform test printing without printing an image on a recording medium, in a case where the inspection result of the reprinted material is fail; and [0120] cancel the printing of the subsequent printed material, in a case where an inspection result of a test printed material by the test printing is pass. [0121] (((3)))

[0122] The information processing system according to (((2))), wherein the processor is configured to: [0123] perform the test printing on a recording medium that is accommodated in the same accommodation unit as an accommodation unit of the recording medium used for the failed printed material. [0124] (((4)))

[0125] The information processing system according to ((2)) or ((3)), wherein the processor is configured to: [0126] in a case where the inspection result of the test printed material is fail, perform retest printing without printing an image on a recording medium that is accommodated in another accommodation unit different from one accommodation unit in which the recording medium used for the test printing is accommodated; and [0127] in a case where an inspection result of a retest printed material by the retest printing is pass, print the subsequent printed material by

using the recording medium accommodated in the other accommodation unit. [0128] (((5)))

[0129] The information processing system according to (((4))), wherein the processor is configured to: [0130] set, as the other accommodation unit, an accommodation unit in which a recording medium having the same size and the same type as the recording medium accommodated in the one accommodation unit is accommodated. [0131] (((6)))

[0132] The information processing system according to ((4)) or ((5)), wherein the processor is configured to: [0133] in a case where the inspection result of the test printed material is fail, perform the retest printing until there are no more other accommodation units. [0134] (((7)))

[0135] The information processing system according to any one of (((1))) to (((6))), wherein the processor is configured to: [0136] discharge a printed material having the inspection result of pass to a pass discharge unit; [0137] discharge the printed material having the inspection result of fail and a printed material being printed following the failed printed material to a fail discharge unit; [0138] discharge the reprinted material having the inspection result of pass to the pass discharge unit; and [0139] discharge the reprinted material having the inspection result of fail to the fail discharge unit. [0140] (((8)))

[0141] The information processing system according to (((7))), wherein the processor is configured to: [0142] discharge the subsequent printed material having an inspection result of pass to the pass discharge unit; and [0143] in a case where the subsequent printed material having the inspection result of fail is generated, discharge the failed subsequent printed material and the printed material being printed following the failed subsequent printed material to the fail discharge unit, and stop the printing of the subsequent printed material. [0144] (((9)))

[0145] The information processing system according to any one of (((1))) to (((8))), wherein the processor is configured to: [0146] determine an inspection result of the subsequent printed material for each printing of the subsequent printed material; and [0147] in a case where a predetermined number of subsequent printed materials having the inspection result of pass are continuous, perform continuous printing of the subsequent printed material following the passed subsequent printed materials. [0148] (((10)))

[0149] An information processing program causing a computer to execute a process comprising: [0150] performing continuous printing of a plurality of printed materials while inspecting printing quality; [0151] stopping, in a case where a printed material having an inspection result of fail is generated, the continuous printing of the printed materials and reprinting the failed printed material; and [0152] printing, in a case where an inspection result of the reprinted printed material is pass, a subsequent printed material following the passed reprinted material. The foregoing description of the exemplary embodiments of the present invention has been provided for the purposes of illustration and description. It is not intended to be exhaustive or to limit the invention to the precise forms disclosed. Obviously, many modifications and variations will be apparent to practitioners skilled in the art. The embodiments were chosen and described in order to best explain the principles of the invention and its practical applications, thereby enabling others skilled in the art to understand the invention for various embodiments and with the various modifications as are suited to the particular use contemplated. It is intended that the scope of the invention be defined by the following claims and their equivalents.

Claims

1. An information processing system comprising: a processor configured to: perform continuous printing of a plurality of printed materials while inspecting printing quality; in a case where a printed material having an inspection result of fail is generated, stop the continuous printing of the printed materials and reprint the failed printed material; and in a case where an inspection result of the reprinted printed material is pass, print a subsequent printed material following the passed reprinted material.

2. The information processing system according to claim 1, wherein the processor is configured to: perform test printing without printing an image on a recording medium, in a case where the inspection result of the reprinted material is fail; and cancel the printing of the subsequent printed material, in a case where an inspection result of a test printed material by the test printing is pass.
3. The information processing system according to claim 2, wherein the processor is configured to: perform the test printing on a recording medium that is accommodated in the same accommodation unit as an accommodation unit of the recording medium used for the failed printed material.
4. The information processing system according to claim 2, wherein the processor is configured to: in a case where the inspection result of the test printed material is fail, perform retest printing without printing an image on a recording medium that is accommodated in another accommodation unit different from one accommodation unit in which the recording medium used for the test printing is accommodated; and in a case where an inspection result of a retest printed material by the retest printing is pass, print the subsequent printed material by using the recording medium accommodated in the other accommodation unit.
5. The information processing system according to claim 4, wherein the processor is configured to: set, as the other accommodation unit, an accommodation unit in which a recording medium having the same size and the same type as the recording medium accommodated in the one accommodation unit is accommodated.
6. The information processing system according to claim 5, wherein the processor is configured to: in a case where the inspection result of the test printed material is fail, perform the retest printing until there are no more other accommodation units.
7. The information processing system according to claim 1, wherein the processor is configured to: discharge a printed material having the inspection result of pass to a pass discharge unit; discharge the printed material having the inspection result of fail and a printed material being printed following the failed printed material to a fail discharge unit; discharge the reprinted material having the inspection result of pass to the pass discharge unit; and discharge the reprinted material having the inspection result of fail to the fail discharge unit.
8. The information processing system according to claim 7, wherein the processor is configured to: discharge the subsequent printed material having an inspection result of pass to the pass discharge unit; and in a case where the subsequent printed material having the inspection result of fail is generated, discharge the failed subsequent printed material and the printed material being printed following the failed subsequent printed material to the fail discharge unit, and stop the printing of the subsequent printed material.
9. The information processing system according to claim 1, wherein the processor is configured to: determine an inspection result of the subsequent printed material for each printing of the subsequent printed material; and in a case where a predetermined number of subsequent printed materials having the inspection result of pass are continuous, perform continuous printing of the subsequent printed material following the passed subsequent printed materials.
10. A non-transitory computer readable medium storing an information processing program causing a computer to execute a process comprising: performing continuous printing of a plurality of printed materials while inspecting printing quality; stopping, in a case where a printed material having an inspection result of fail is generated, the continuous printing of the printed materials and reprinting the failed printed material; and printing, in a case where an inspection result of the reprinted printed material is pass, a subsequent printed material following the passed reprinted material.
11. An information processing method comprising: performing continuous printing of a plurality of printed materials while inspecting printing quality; stopping, in a case where a printed material having an inspection result of fail is generated, the continuous printing of the printed materials and reprinting the failed printed material; and printing, in a case where an inspection result of the

reprinted printed material is pass, a subsequent printed material following the passed reprinted material.
